A QUASI-COMPACT IMMERSION WHOSE PREIMAGE OF A PRINCIPAL OPEN IS AFFINE AND WHOSE COMPLEMENT PIECE OVER V(F) IS CLOSED IS AN AFFINE SCHEME

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Allowed standard prerequisites (editorial).

- The get-affine criterion for immersions (morphisms chapter, Lemma morphisms-lemma-get-affine, canonical tag 0C3A). cited at proof lines 2994-2995. One route to the result, acknowledged up front; the direct proof follows.
- A scheme whose global sections embed it as a quasi-compact open in the spectrum of those sections is quasi-affine (properties chapter, Lemma properties-lemma-quasi-affine, canonical tag 01P9). cited at proof lines 2996-2998. Gives the quasi-compact open immersion j' : V to Spec(A') used to replace A by A' = Gamma(V, O_V).
- A subscheme of an affine scheme isomorphic to a closed subscheme of another affine scheme via an immersion is closed (schemes chapter, Lemma schemes-lemma-section-immersion, canonical tag 01KT). cited at proof lines 3000-3003. Shows j'(V) intersect V(f') is closed in Spec(A') because it maps isomorphically to the closed j(V) intersect V(f).
- If the preimage of D(f) is affine then j maps it isomorphically onto D(f), and conversely an affine open mapping isomorphically onto its image inside an affine open identifies with a principal open there (properties chapter, Lemma properties-lemma-invert-f-affine, canonical tag 01P8). cited at proof lines 3011-3012 and 3015-3017. First use: j maps j^{-1}D(f) isomorphically onto D(f), so D(g) meets V(f); second use: D(g) is isomorphic to j^{-1}D(g) inside j^{-1}D(f), forcing D(g) inside D(f) and giving the contradiction.

Source reference. Original lemma 0C3A: sources/stacks-chow/tags.txt, source lines 15185–15185. Cited at proof lines 2994-2995: the morphisms-chapter criterion from which the result also follows; the direct proof is given alongside.

Source reference. Original lemma 01P9: sources/stacks-chow/tags.txt, source lines 2222–2222. Cited at proof lines 2996-2998: j' : V to Spec(Gamma(V, O_V)) is a quasi-compact open immersion.

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Source reference. Original lemma 01KT: sources/stacks-chow/tags.txt, source lines 2062–2062. Cited at proof lines 3000-3003: j'(V) intersect V(f') is closed in Spec(A') since it maps isomorphically to the closed j(V) intersect V(f).

Source reference. Original lemma 01P8: sources/stacks-chow/tags.txt, source lines 2221–2221. Cited at proof lines 3011-3012 (j maps j^{-1}D(f) isomorphically onto D(f)) and 3015-3017 (D(g) isomorphic to j^{-1}D(g) inside j^{-1}D(f), forcing D(g) inside D(f)).

Lemma 0.2. *Let $j : V \to \mathrm{Spec}(A)$ be a quasi-compact immersion of schemes. Let $f \in A$ be such that $j^{-1}D(f)$ is affine and $j(V) \cap V(f)$ is closed. Then V is affine.*

Proof. This follows from Morphisms, Lemma 0C3A but we will also give a direct proof. Let $A' = \Gamma(V, \mathcal{O}_V)$. Then $j' : V \to \mathrm{Spec}(A')$ is a quasi-compact open immersion, see Properties, Lemma 01P9. Let $f' \in A'$ be the image of f. Then $(j')^{-1}D(f') = j^{-1}D(f)$ is affine. On the other hand, $j'(V) \cap V(f')$ is a subscheme of $\mathrm{Spec}(A')$ which maps isomorphically to the closed subscheme $j(V) \cap V(f)$ of $\mathrm{Spec}(A)$. Hence it is closed in $\mathrm{Spec}(A')$ for example by Schemes, Lemma 01KT. Thus we may replace A by A' and assume that j is an open immersion and $A = \Gamma(V, \mathcal{O}_V)$.

In this case we claim that $j(V) = \mathrm{Spec}(A)$ which finishes the proof. If not, then we can find a principal affine open $D(g) \subset \mathrm{Spec}(A)$ which meets the complement and avoids the closed subset $j(V) \cap V(f)$. Note that j maps $j^{-1}D(f)$ isomorphically onto $D(f)$, see Properties, Lemma 01P8. Hence $D(g)$ meets $V(f)$. On the other hand, $j^{-1}D(g)$ is a principal open of the affine open $j^{-1}D(f)$ hence affine. Hence by Properties, Lemma 01P8 again we see that $D(g)$ is isomorphic to $j^{-1}D(g) \subset j^{-1}D(f)$ which implies that $D(g) \subset D(f)$. This contradiction finishes the proof. $\square$